

Recommendations Report (Part D)

Introduction

This report presents recommendations following the Part C evaluation of two candidate models, a Logistic Regression baseline (Model 1) and a Random Forest ensemble (Model 2), built on the public UCI Bank Marketing dataset (Moro, Cortez and Rita, 2014) as a proof of concept for STADIOEquities' account-activation problem. It addresses which model performed best, how it can be improved, how it will need to be adapted for STADIOEquities' own data once received from Study Session 1 (SS1), and how these results compare with the literature reviewed in Part A.

1. Which model is best?

On the held-out test set, the two models achieved comparable discriminative power, a ROC AUC of 0.802 for Logistic Regression and 0.806 for Random Forest. A paired bootstrap comparison confirmed this small gap is not statistically significant, so overall ranking ability is effectively tied.

The models differ, however, in the trade-off they make. Random Forest achieved materially higher precision (0.459 vs 0.366), a higher F1-score (0.524 vs 0.468) and higher accuracy (87.5% vs 83.4%), while Logistic Regression retained higher recall (0.650 vs 0.612), meaning it missed fewer genuine activators at the cost of flagging more accounts that would not ultimately activate.

Because STADIOEquities' stated goal is to move from blanket onboarding communications to targeted, individualised nudges, minimising wasted outreach matters as much as catching every activator. We therefore recommend **Random Forest (Model 2) as the primary model**, since its higher precision means a larger share of the accounts it flags will genuinely be worth the intervention effort.

Logistic Regression should be retained as a secondary, interpretable benchmark since its coefficients can be explained directly to non-technical stakeholders and used to check Model 2's behaviour.

2. How can the selected model be improved?

Four improvements are recommended before Random Forest is used operationally. First, systematic hyperparameter tuning (over the number of trees, tree depth and minimum leaf size) was not performed here and should be, since near-default settings were used throughout Part B.

Second, its predicted probabilities should be calibrated, since Random Forest probabilities are known to be poorly calibrated out of the box, which matters if a cost-based decision threshold is later used to decide who receives a nudge.

Third, gradient-boosting alternatives (e.g. XGBoost) should be benchmarked against it, given that Yoon, Zame and van der Schaar (2018) found ensemble methods of this kind to be strong performers on this dataset.

Lastly, the current class-imbalance handling should be compared against alternatives such as SMOTE, and validated against STADIOEquities' own class balance rather than reused unchanged.

3. How will the selected model need to be adapted to work with the SS1 data?

Several adaptations will be required before Model 2 can be applied to STADIOEquities' data. The bank-specific features (job, marital status, prior campaign contacts) will need to be replaced with equivalents drawn from the account/registration, behavioural, demographic and marketing extracts. Notably, the macro-economic indicators (like the Euribor rate) were among the most predictive features here, but have no equivalent in STADIOEquities' data, they should be replaced with engineered behavioural features such as time-to-first-deposit, onboarding-step completion, session activity and acquisition channel, which the project's own risk log identifies as the factors the drop-off clusters by.

The target variable will also need to be redefined, rather than a single marketing-call outcome, it should be a binary funded/not-funded label measured within a fixed post-registration window (e.g. 90 days), built only from data available at an early checkpoint (e.g. day 7 or 14) to avoid leakage. The same principle that led us to exclude call duration in Part B.

This depends on first validating the account-ID join across the four extracts, a risk already flagged for this project.

Finally, because STADIOEquities' own activation rate (roughly 59%, per the business briefing) is far less imbalanced than the ~11% positive rate in the public dataset, the imbalance handling and decision threshold will need to be re-tuned rather than carried over, and validation should use an out-of-time split by registration cohort, since our testing showed performance is sensitive to temporal drift.

4. Do your results align with the related literature from Part A?

Yes. Moro, Cortez and Rita (2014) report an AUC of 0.80 for their best model under a realistic, time-based evaluation, closely matching both our Logistic Regression (0.802) and Random Forest (0.806) results on a random split, and consistent with the drop to 0.70–0.73 we observed once both models were re-tested on a time-ordered split similar in spirit to their own evaluation design.

The modest advantage of the ensemble method found here (Random Forest outperforming Logistic Regression on precision, F1-score and out-of-time robustness) is also consistent with the wider literature: Liang et al. (2024) found non-linear, autoencoder-based feature compression outperformed linear methods such as PCA on this exact dataset, and Yoon, Zame and van der Schaar (2018) found a heterogeneous ensemble of predictors outperformed single global models, including XGBoost, on the same data.

Together, these results support the Part A conclusion that ensemble or non-linear approaches are the more promising direction for STADIOEquities' own modelling, while confirming that a simple linear baseline remains a valuable, interpretable point of reference.

Conclusion

We recommend progressing with the Random Forest as the lead model, subject to hyperparameter tuning, probability calibration and out-of-time validation and re-engineering its features specifically for STADIOEquities' onboarding and behavioural data once the SS1 extracts are received. These findings are consistent with the wider academic literature and support proceeding to the next phase with confidence in the modelling approach taken.

References

Liang, Y., Li, X., Huang, X., Zhang, Z. and Yao, Y. (2024) 'An automated data mining framework using autoencoders for feature extraction and dimensionality reduction', *arXiv preprint* arXiv:2412.02211.

Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', *Decision Support Systems*, 62, pp. 22–31.

Yoon, J., Zame, W.R. and van der Schaar, M. (2018) 'ToPs: ensemble learning with trees of predictors', *IEEE Transactions on Signal Processing*, 66(4), pp. 939–948.